

Problem 6

Problem 6

Problem 1 Consider the parabola $f(x) = ax^2 + bx + c$ where a, b, c are constants. For what values of a, b, c is f concave up? For what values of a, b, c is f concave down?

Explanation.

$$\begin{aligned}f'(x) &= 2ax + b \\f''(x) &= 2a\end{aligned}$$

$\implies$ the sign of a will determine if f is concave up or down. When $a < 0$, f is concave down. When $a > 0$, f is concave up. Note: If $a = 0$, f has no concavity because it is a linear function.